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Problem Definition: How to sift through Scientific Paper in a more efficient way?

Definition of the task: Abstractive Summarization of Scientific Paper Using Transformer Based Model.

Input of the Model: Abstract of a Scientific Paper

Output of the Model: A more concise and vocabulary rich summary

Metric: Rouge Score

Dataset: https://huggingface.co/datasets/scillm/scientific_papers-archive

TF-IDF: TF-IDF is a way to measure the importance of each word in a corpus, it works by calculating TF (how often the word appears in a document) and IDF (how rare the word appears in the corpus). Then TF and IDF is multiplied to get the importance value.

Formula:

$$TF(t, d) = \frac{f_t}{n_d}$$

$$IDF(t, D) = \log\left(\frac{N}{|d \in D: t \in d|}\right)$$

$$TF - IDF(t, d, D) = TF(t, d) \times IDF(t, D)$$

Jokes:

Why did the data miner break up with the text document?

Because it had too much *baggage* and not enough *context*! 😊